

# Wiring presence into the face

**What this is.** The presence layer - blink, gaze, breath, drift - now runs in Unreal as `UAmandaPresenceComponent`, ticking on the character and subscribed to the orchestrator's bridge. Its values are correct and published. Nothing reads them yet. This is the join that puts them on the face.

**What it is not.** It does not touch the mouth. The mouth is solved from audio by MetaHuman's own speech solver, and presence has no business overriding it. Everything below layers *after* the Live Link pose and above the eyes.

**Where.** `Content/Amanda/ABP_AmandaFace` - the project's own copy. Do not do this in `/Game/MetaHumans/Common/Animation/ABP_MH_LiveLink`: that tree is regenerated by `assemble_metahuman.py` and gitignored, so the work would be destroyed by the next character rebuild.

## Before you start

Open `ABP_AmandaFace`. You should see an **AnimGraph** and an **EventGraph**, and in the Variables panel the ones Epic ships: `LLink_Face_Subj`, `ARKit_HeadRotation`, `LLink_Face_Head`, `CustomControlValues`, `JawOpenAlpha`.

The presence component is already attached to `BP_MHC_Seo` - you do not need to add it. If you ever rebuild the character, re-run:

```
Scripts/add_presence_component.py
Scripts/bind_face_blueprint.py
```

## Part A - EventGraph: read the presence values

The animation blueprint needs a reference to the component, once, and then a read per frame.

**A1. Add two variables** (Variables panel, +):

| Name     | Type                                                  |
|----------|-------------------------------------------------------|
| Presence | Amanda Presence Component ( <i>object reference</i> ) |
| Face     | Amanda Face State ( <i>structure</i> )                |

**A2. Cache the component. On Event Blueprint Initialize Animation:**

```
Event Blueprint Initialize Animation
-> Get Owning Actor
-> Get Component by Class    (Component Class = Amanda Presence Component)
-> SET Presence
```

`Get Owning Actor` is on the anim instance - do not use *Try Get Pawn Owner*, the MetaHuman is not a pawn.

**A3. Read it each frame. On Event Blueprint Update Animation:**

```
Event Blueprint Update Animation
-> Get Presence -> Is Valid ? (branch; skip everything if false)
-> Get Presence -> Get Face State
-> SET Face
```

Guarding on `IsValid` matters: the anim blueprint also runs in the asset preview window, where there is no actor and no component, and an unguarded read throws every frame.

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## Part B - AnimGraph: put the values on the face

Find the existing chain that ends at **Output Pose**. The Live Link Pose node feeds it. You are inserting nodes *between* that and the output.

### B1. Eyelids - the one that matters most

Add a **Modify Curve** node after the Live Link pose.

- Set **Apply Mode** to **Max** for blink curves. Blink should win over

whatever the audio solver is doing to the eyes, and Max means presence can close the lid but never force it open.

- Right-click the node -> **Add Curve Pin**. In the name box type `blink`. The

dropdown is populated from the face skeleton, so **pick the names it offers** rather than typing from memory - they differ between MetaHuman versions. Expect a left and a right.

Feed both pins from:

```
Break Amanda Face State (from Face)
-> Lid Open    -> [1.0 minus this]    -> curve pins
```

LidOpen is 1.0 open and 0.0 closed; a blink *curve* is the other way round, so subtract from one. If the eyes end up permanently shut, this is why.

### B2. Head - drift and gaze lag

The head has a variable Epic already wired: `ARKit_HeadRotation`.

In the **EventGraph**, in the same Update Animation chain as A3:

```
Get Presence -> Get Head Rotation    -> SET ARKit_HeadRotation
Set LLink_Face_Head = false
```

`GetHeadRotation` is a helper on the component that packs pitch/yaw/roll into a rotator. Setting `LLink_Face_Head` false stops the Live Link source also claiming the head - otherwise the two fight and the result is a twitch.

Values are **degrees, and small**: drift is under a degree, gaze lag a few degrees. If nothing appears to happen, that is expected - check the numbers move before assuming it is broken.

### B3. Eye aim

Try `CustomControlValues` first - it is a Name -> float map the graph already consumes:

```
Break Amanda Face State
-> Eye Yaw, Eye Pitch -> normalise to -1..1 (divide by ~15)
-> Map Add to CustomControlValues under the eye-look control names
```

Use the same trick as B1 to find the names: type `eyeLook` and take what the dropdown offers.

If the map turns out not to reach the eye controls, fall back to a **Look At** node on the eye bones, driven from the same two values. That is more work and worth avoiding unless needed.

## B4. Breath

Lowest priority, and it belongs on the **body**, not the face - ABP\_Body\_PostProcess, chest and shoulders. Breath is 0.0 empty to 1.0 full. Leave it until the face is right.

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## Order matters

```
Live Link Pose -> Modify Curve (blink) -> head/eye -> Output Pose
```

Presence goes **after** Live Link so it layers on top. Put it before and the solver overwrites the blink every frame, which looks exactly like the component not working.

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## Verifying

Compile, save, then **Simulate** (not just the viewport - an animation blueprint does not evaluate in the editor viewport, which cost an afternoon here).

She should blink irregularly, roughly every three to six seconds, with the eyes drifting slightly between fixations and the head moving under a degree.

To check the numbers rather than your eyes:

```
UnrealEditor.exe Amanda.uproject /Game/Amanda/Maps/LookDev ^  
-ExecCmds="py <scratch>\sample_presence.py"
```

Last measured, over 40 seconds with no orchestrator attached: 14 blinks (21/min), lid range 0.000-1.000, eye yaw -12.3 deg to +11.9 deg, all five gaze targets visited.

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## If it does not work

**Nothing moves at all.** Check you are in Simulate or Play, not the viewport. Then check Presence is valid - if Get Component by Class returns null, the component is on the Blueprint but you are previewing the anim blueprint in isolation, where there is no actor.

**The eyes are shut.** B1, inverted. LidOpen 1.0 means *open*.

**It works, then stops after a rebuild.** BP\_MHC\_Seo lives in the regenerated tree. Re-run add\_presence\_component.py and bind\_face\_blueprint.py.

**The head twitches.** Both presence and Live Link are claiming it. LLink\_Face\_Head should be false.

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## What the values mean

| Field             | Range                  | Notes                                          |
|-------------------|------------------------|------------------------------------------------|
| LidOpen           | 1.0 open -> 0.0 closed | Blink curves usually want the inverse          |
| EyeYaw / EyePitch | +/-16 deg              | Includes micro-saccades; never perfectly still |

|                                |            |                                                     |
|--------------------------------|------------|-----------------------------------------------------|
| HeadYaw / HeadPitch / HeadRoll | +/-3 deg   | Gaze lag plus drift. Deliberately tiny              |
| Breath                         | 0.0 -> 1.0 | Chest, not face                                     |
| Preset / Intensity             | -          | From the performance director, when one is attached |
| GazeTarget                     | enum       | Diagnostics                                         |

The reasoning behind every constant is in `avatar-orchestrator/src/amanda/presence/schedulers.py`, which is the reference implementation and has the tests. If behaviour needs changing, change it there and port, so the two do not drift apart.